Ambient-Noise Interferometry Across the Yakima Fold and Thrust Belt: Surface-Wave Modes and Shallow Velocity Heterogeneity

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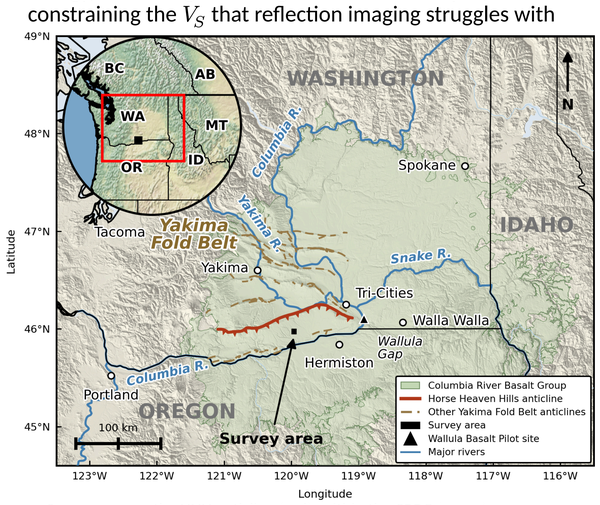
MOTIVATION

- Horse Heaven Hills (HHH): anticlinal ridge of the Yakima Fold and Thrust Belt
- Layered Columbia River Basalt Group (CRBG): dense flow interiors, weathered low-velocity flow tops
- Slow–fast layering favors higher surface-wave modes
- **Project TrapRock**: Carbon Containment Lab and Washington Geological Survey assess the CRBG for carbon storage
- **Low frequencies**: layered basalt scatters high-frequency energy, while long wavelengths reach 1 km depth, constraining the V_S that reflection imaging struggles with

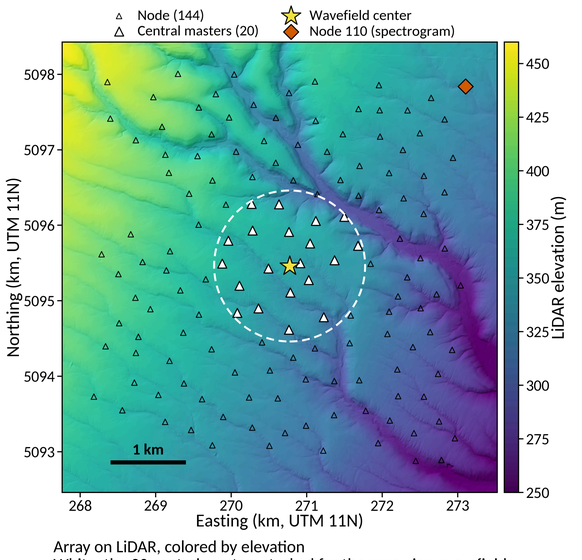


Survey area on the HHH anticline (red), within the CRBG (green)

CONTINUOUS NODAL ARRAY

- **144** three-component (3-C) 2 Hz nodes
- **Seven weeks** continuous, Aug to Oct 2025
- **5.5 km** aperture; **170 m** relief, high NW to low SE
- Passive component of the TrapRock nodal dataset (Girard et al., submitted)

GTI NuSeis D3 node prior to burial



Array on LiDAR, colored by elevation
White: the 20 central masters stacked for the map-view wavefields

REFERENCES AND ACKNOWLEDGMENTS

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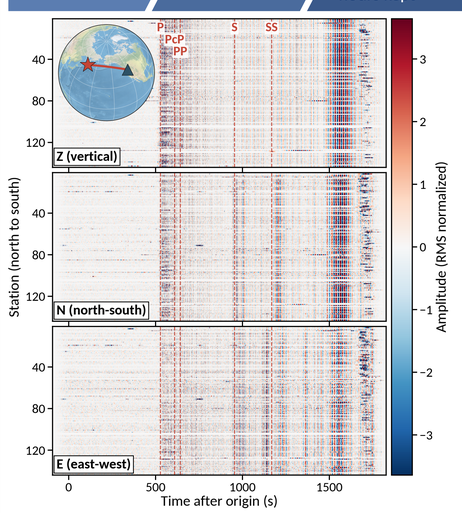
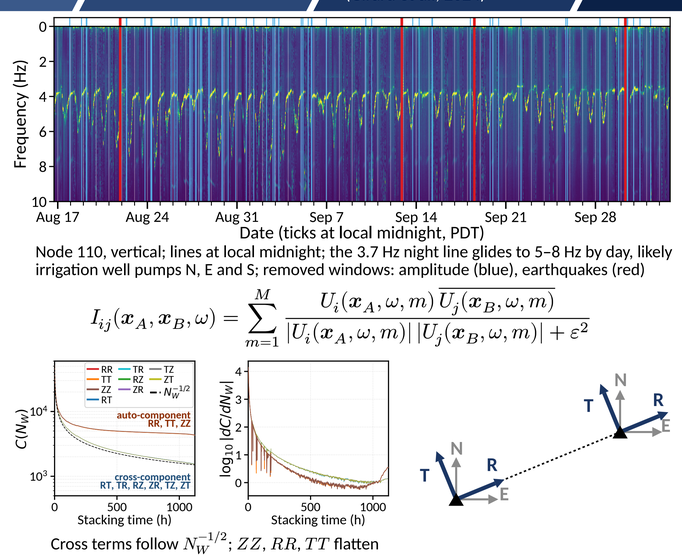
PROCESSING

Decimate anti-alias to 25 Hz → **Detrend** multi-pass spline → **Window selection** amplitude and earthquakes: **1125 of 1200** hours kept → **Temporal balance** Girard & Shragge (2020) → **GPU cross-coherence** with node locations, rotates N–E to R–T in one pass (Girard et al., 2024) → **9-C VSGs**

virtual shot gathers, 144 × 144 nodes



M_w 7.8 Kamchatka aftershock, 18 Sep 2025: coherent on all 144 stations and removed before stacking



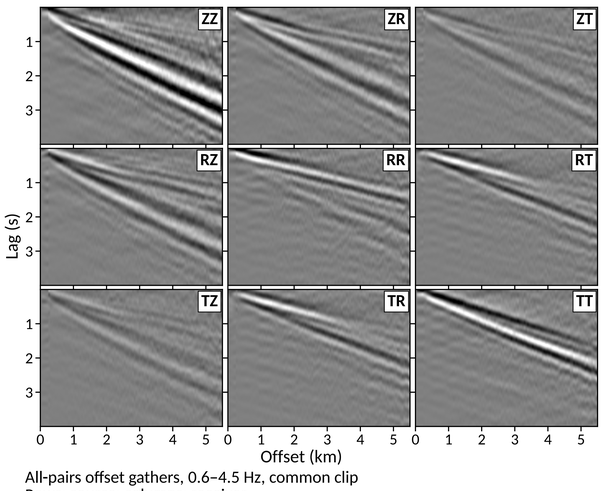
Node 110, vertical; lines at local midnight; the 3.7 Hz night line glides to 5–8 Hz by day, likely irrigation well pumps N, E and S; removed windows: amplitude (blue), earthquakes (red)

$$I_{ij}(\boldsymbol{x}_A, \boldsymbol{x}_B, \omega) = \sum_{m=1}^{M} \frac{U_i(\boldsymbol{x}_A, \omega, m)\, \overline{U_j(\boldsymbol{x}_B, \omega, m)}}{|U_i(\boldsymbol{x}_A, \omega, m)|\, |U_j(\boldsymbol{x}_B, \omega, m)| + \varepsilon^2}$$

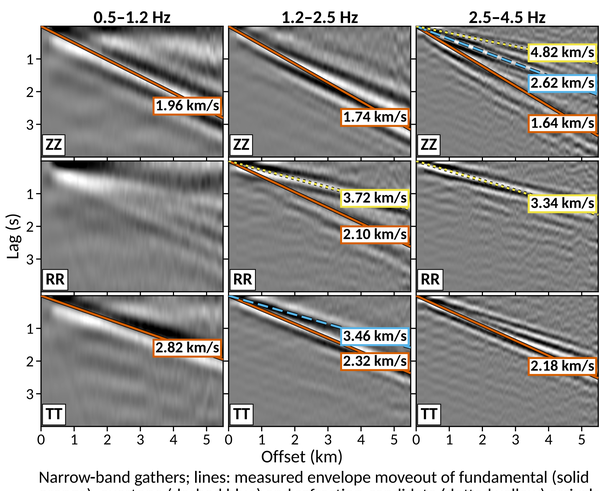
Cross terms follow $N_W^{-1/2}$; ZZ, RR, TT flatten

NINE-COMPONENT VSGs

- Surface-wave-dominated field
- ZZ, RR (Rayleigh) and TT (Love): dispersive moveout
- ZR, RZ: energetic, reciprocity-linked
- ZT, TZ, RT, TR: weak but nonzero, indicating heterogeneity



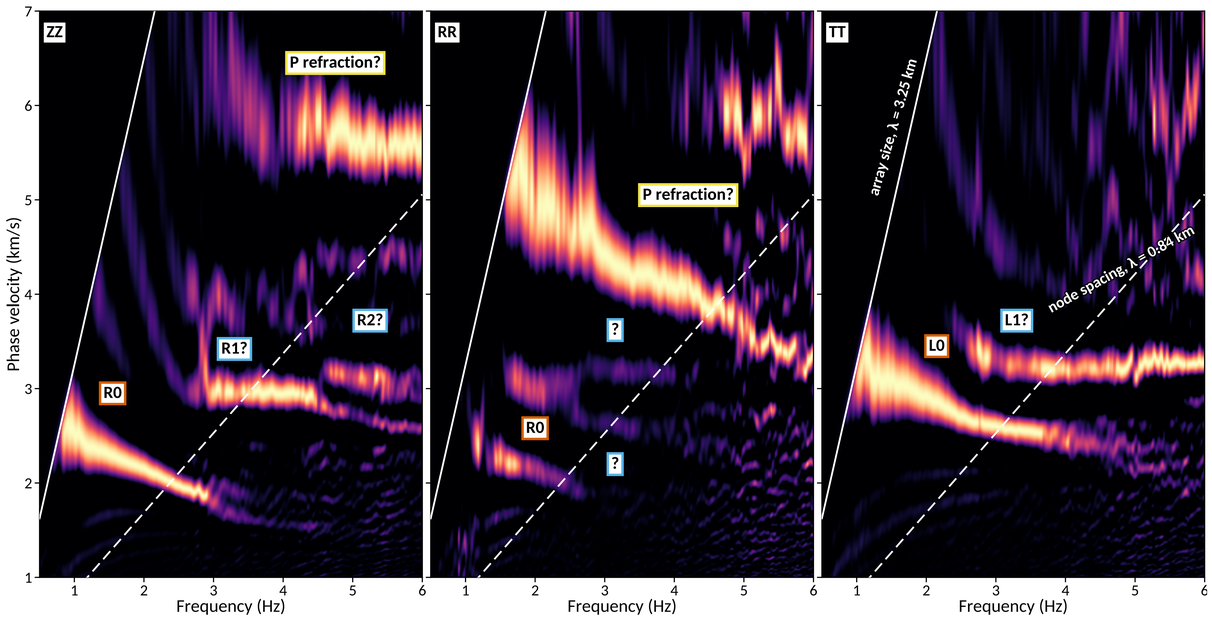
All-pairs offset gathers, 0.6–4.5 Hz, common clip
Rows: source; columns: receiver



Narrow-band gathers; lines: measured envelope moveout of fundamental (solid orange), overtone (dashed blue) and refraction-candidate (dotted yellow) arrivals

DISPERSION AND MODES

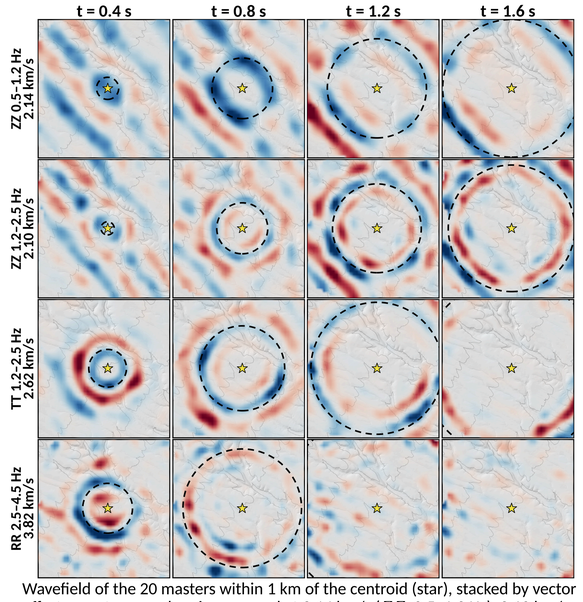
- Normal dispersion on Rayleigh and Love: V_S increases with depth
- At least one higher-mode Rayleigh branch on RR
- Love slightly faster than Rayleigh



Frequency-Bessel stacks of the all-pairs gathers, scaled per frequency; branches marked "?" are candidates; solid white line: array-size limit, λ = 3.25 km, longer wavelengths masked; dashed white line: node-spacing limit, λ = 0.84 km, shorter wavelengths can alias

TOPOGRAPHY VS. HETEROGENEITY

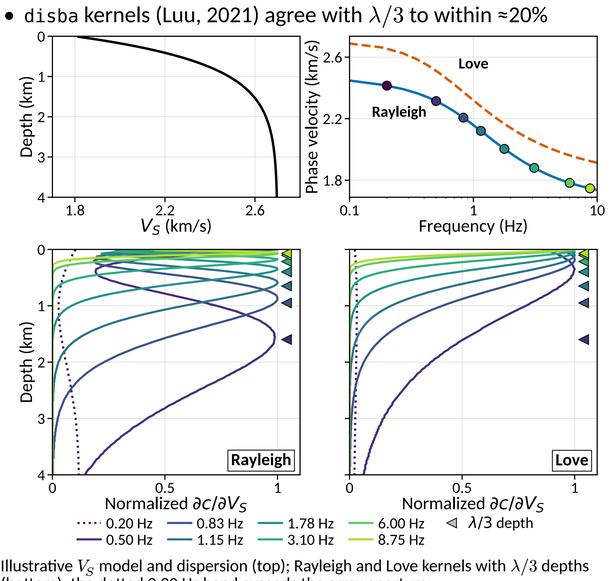
- Wavelengths **0.5–3.5 km** ≫ **170 m** relief: topography is sub-wavelength
- Wavefronts cross the array undeflected
- Rayleigh–Love cross-term leakage rises from **0.34** (0.1–0.3 Hz) to **0.90** (5–8 Hz)
- Departures from 1-D: lateral velocity heterogeneity in the shallow section
- Linear bands: Rayleigh energy from the SW, 2.2–2.5 km/s, confirmed by beamforming



Wavefield of the 20 masters within 1 km of the centroid (star), stacked by vector offset, over topography; rings expand at 2.14 km/s (ZZ, 0.5–1.2 Hz), 2.10 km/s (ZZ, 1.2–2.5 Hz), 2.62 km/s (TT) and 3.82 km/s (RR); dashed: measured crest

DEPTH SENSITIVITY

- Wavelength scaling: $z \approx \lambda/3 = c/(3f)$
- Sensitivity from **≈70 m to 1 km**
- `disba` kernels (Luu, 2021) agree with $\lambda/3$ to within ≈20%



Illustrative V_S model and dispersion (top); Rayleigh and Love kernels with $\lambda/3$ depths (bottom); the dotted 0.20 Hz band exceeds the array aperture

CONCLUSIONS

- Surface-wave-dominated 9-C field: Rayleigh on $\{ZZ, ZR, RZ, RR\}$, Love on TT
- Normal dispersion, with at least one higher mode on both Rayleigh and Love
- Sensitivity from 70 m to 1 km depth
- Topography is sub-wavelength; departures from 1-D record shallow lateral velocity heterogeneity

FUTURE WORK

- Rayleigh + Love branches: constrain a fully elastic model
- Ambient wavefield tomography (AWT) and eikonal tomography for lateral structure
- I-FDG (Davis et al., 2026, *GJI*) and HVSR for vertical constraints
- Higher modes add constraints to AWT and I-FDG
- Trans-dimensional Bayesian inversion: AWT, eikonal, I-FDG and HVSR constrain each other, giving $V_S(z)$ with uncertainty
- Pre-injection baseline for time-lapse monitoring